Shower Device

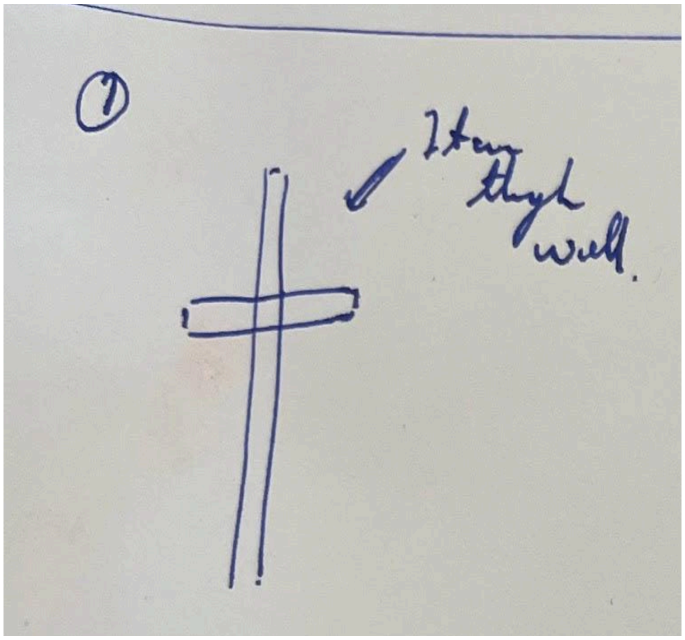
Happy Campers

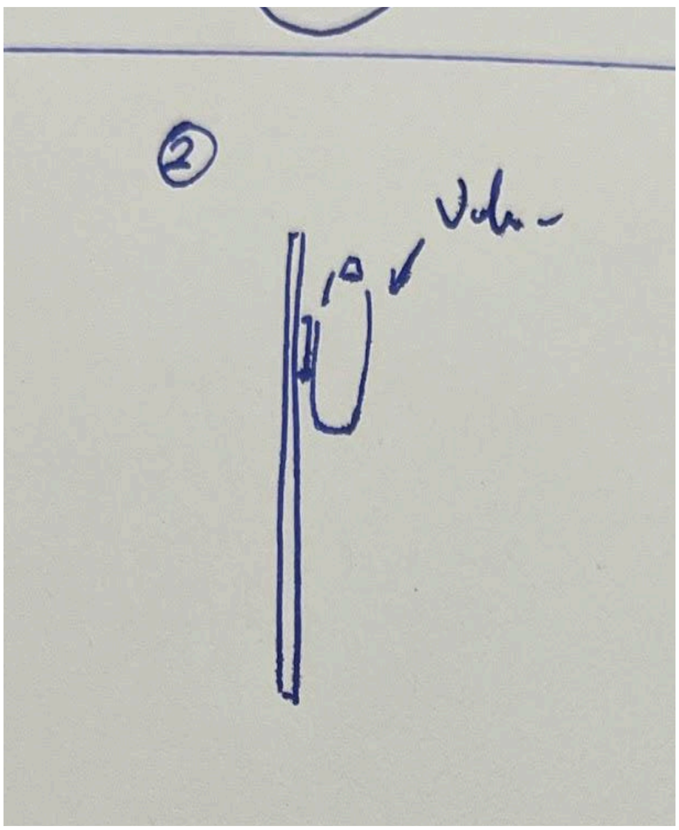
Element D: Design Concept Generation, Analysis, and Selection

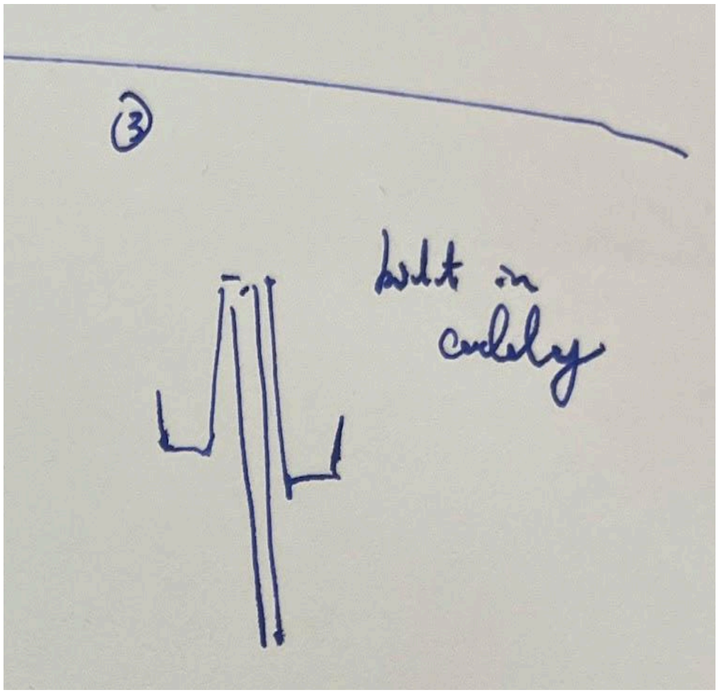
Dormitory residents and dormitory guests of our boarding school need a device to place their shower products. This is because they would want their shower products to be easily accessible in the communal shower. They would also like to do this without having a personal shower caddy and without setting it on the floor because it is unsanitary. The main stakeholders are the dormitory residents. They need a solution because the showers in the dormitory were not built with a solution to this problem and dormitory residents are not permitted to leave their shower products in the shower area. The only place to put a device in the dormitory showers are on a relatively narrow divider wall of less than an inch wide.

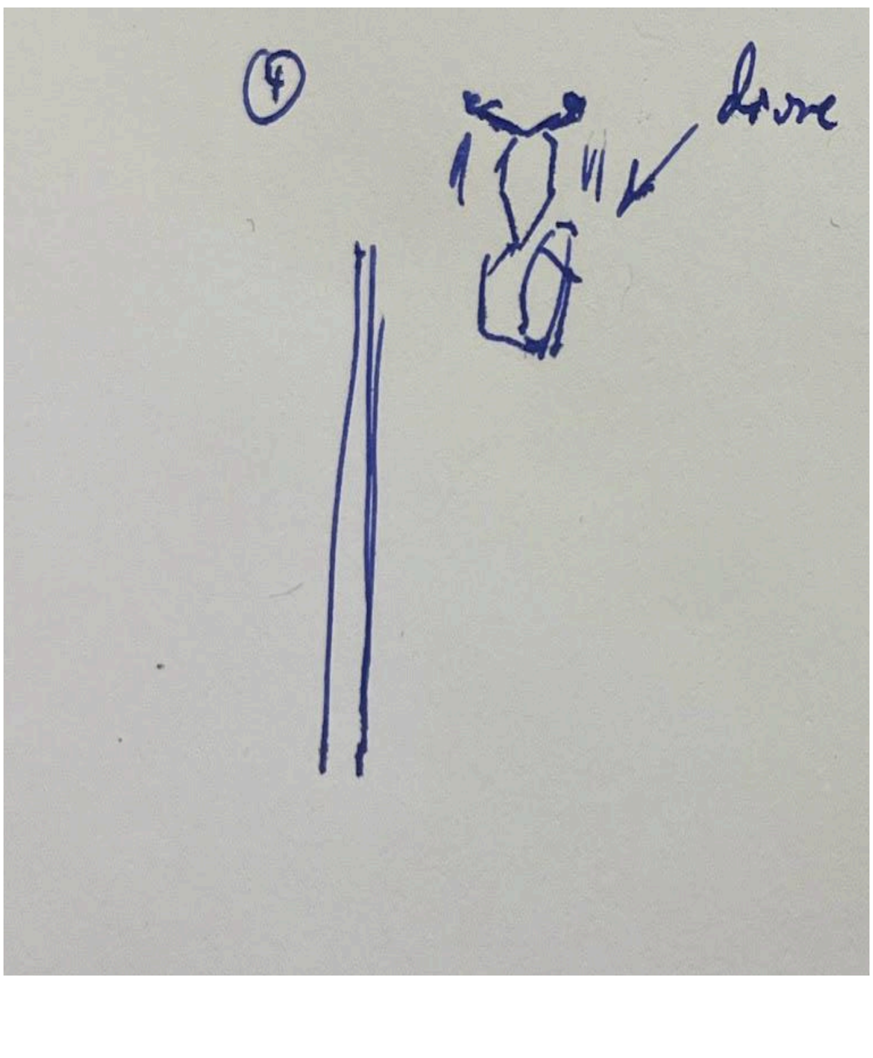
The team brainstormed a variety of ideas that have the potential to fulfil the stakeholder's need individually for approximately ten minutes, then shared with team members. A few new ideas are generated during the discussion process. Additionally, the team determined the basic dimensions of the design were finalised according to the dimensions of the common shower caddy in order to guarantee that the shower caddy can be placed on the design.

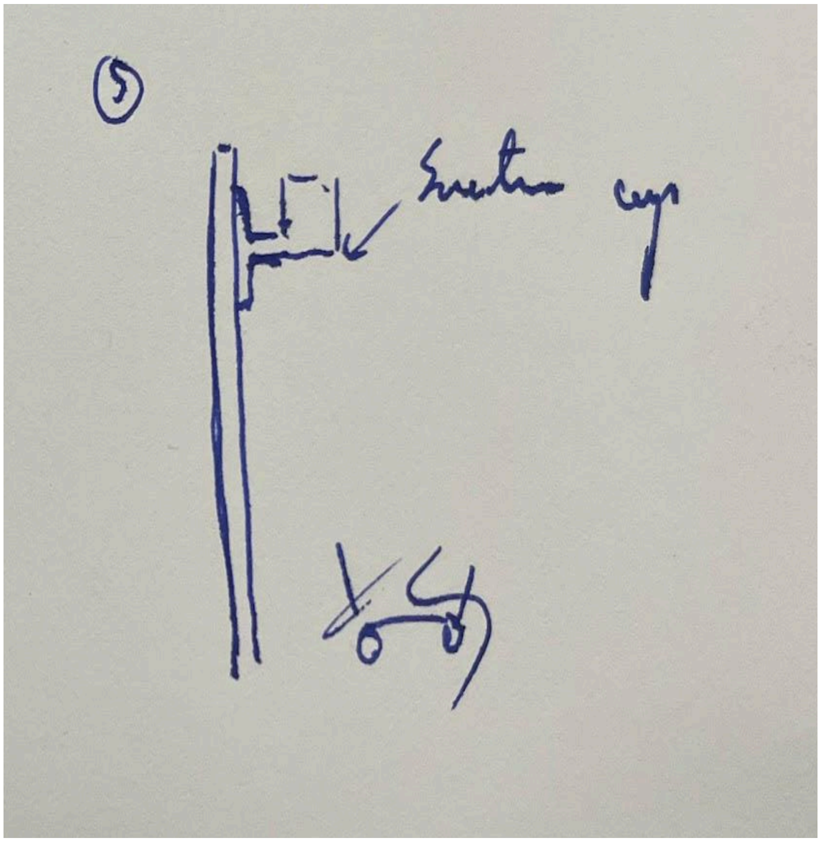
The team combined the ideas from brainstorming and formed five distinct solutions


1
Steam through wall.


2
Valve


3
bolts in
carefully


4
drive



, and then compared these five designs with the current solution The team included the following conditions in the decision matrix.

Team 2: Happy Campers

	Proposed Solutions					
Design Requirements	**Current Solution: Caddy that sits on wall**	**Idea #1: One Piece Through Hole**	**Idea #2: Velcro**	**Idea #3: Double Sided**	**Idea #4: Stabilizing Drone**	**Idea #5: Suction Cup**
Waterproof	0	0	0	0	-1	0
Sturdy - holds up to 10lb	0	1	-1	0	1	-1
Elevated off the floor	0	0	0	0	1	0
Fits 5 bottles and a razor or an external shower caddy	0	1	-1	0	0	0
Easily cleanable with standard cleaning supplies	0	1	-1	0	-1	-1
No standing water	0	-1	1	0	1	0
Balanced/Stable	0	0	-1	0	1	-1
Price	0	0	1	0	-1	0
Should be small <6in deep, <12in wide	0	0	1	0	-1	0
Feasible	0	-1	-1	-1	-2	-1
Total Score	**0**	**1**	**-1**	**-1**	**-1**	**-3**

Scoring
0 = same as current solution

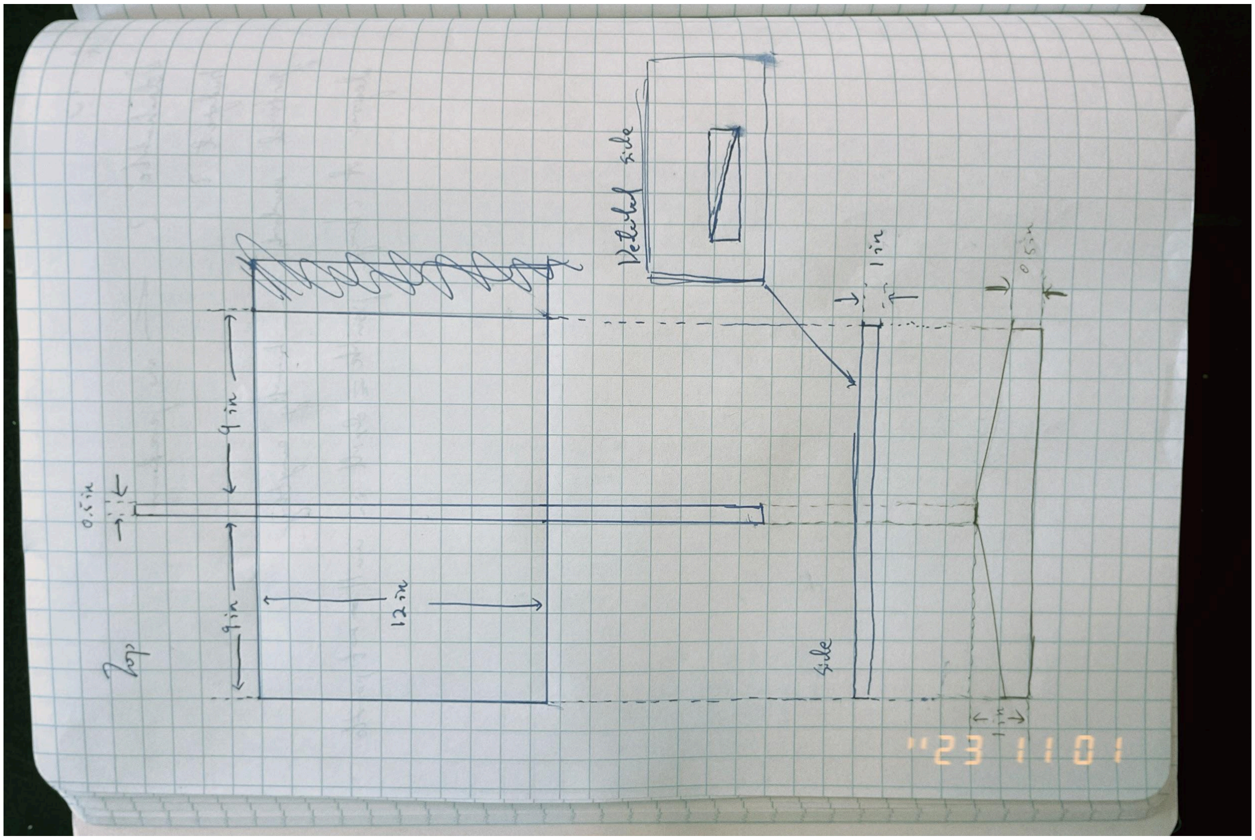
By comparing the tem mentioned properties, the ‘One piece though wall’ idea was considered to be the most applicable and will have the best performance, and was selected by the team.

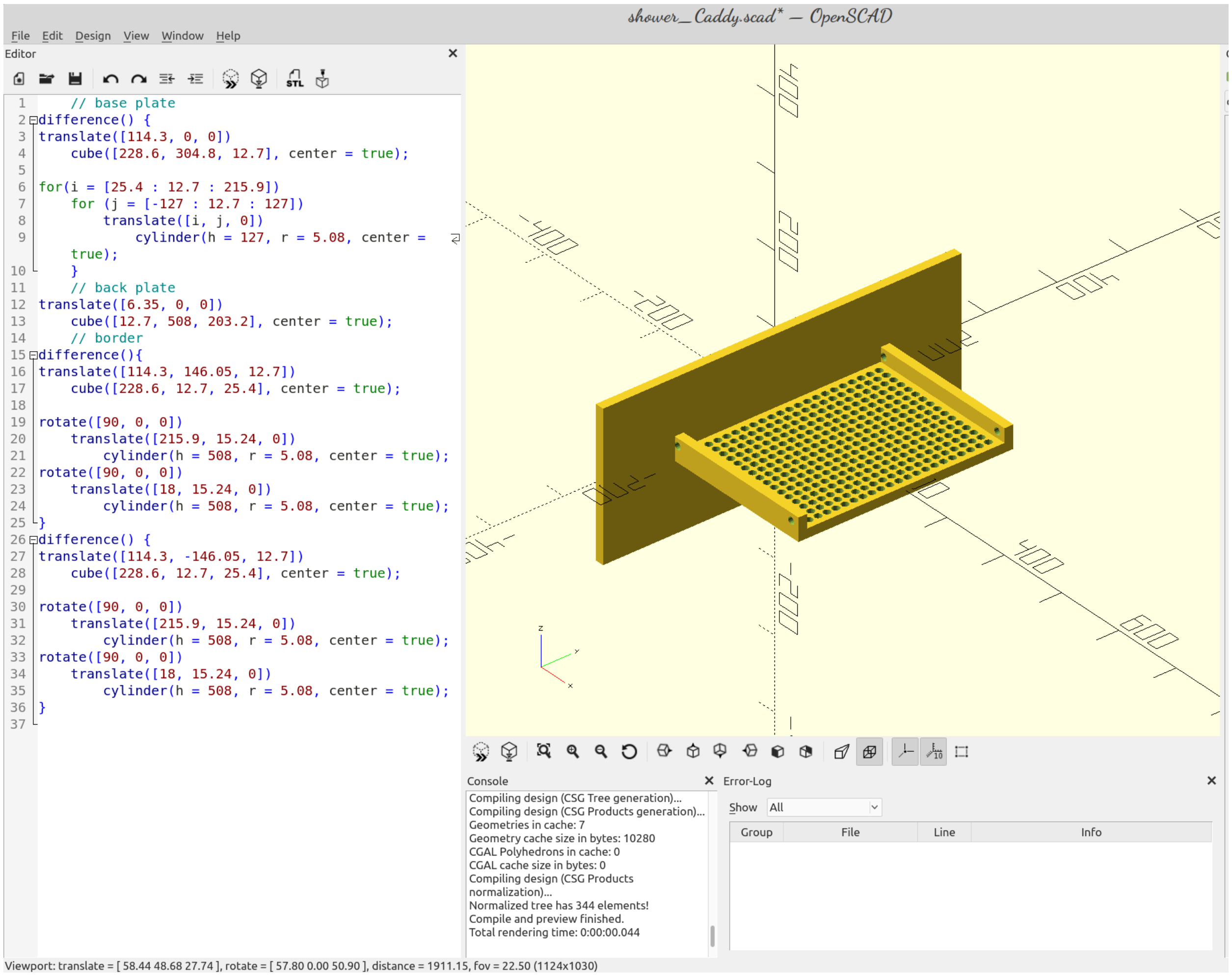
The piece will be constructed out using the 3D printer with PLA filament, and it will be mounted in the divider using a hole cut out and will be held in place with flex seal.

A few changes are made after the design-a-thon including:

- The piece will not go through the wall, instead it will be suspended with nylon ropes.
- The piece will use an array of holes to drain water instead of the complex structure.
- Side borders are added to keep items placed above in place.
- The new sketch is modelled with OpenSCAD.

The construction will be heavily on Computer Aid Design (CAD). The team selected to use PLA filament because it is more water resistant than PVA, and is relatively cheaper than ABS. The estimated cost will be around $30.




shower_Caddy.scad* — OpenSCAD
File Edit Design View Window Help
Editor
// base plate
difference() {
translate([114.3, 0, 0])
cube([228.6, 304.8, 12.7], center = true);
for(i = [25.4 : 12.7 : 215.9])
for (j = [-127 : 12.7 : 127])
translate([i, j, 0])
cylinder(h = 127, r = 5.08, center = true);
}
// back plate
translate([6.35, 0, 0])
cube([12.7, 508, 203.2], center = true);
// border
difference(){
translate([114.3, 146.05, 12.7])
cube([228.6, 12.7, 25.4], center = true);
rotate([90, 0, 0])
translate([215.9, 15.24, 0])
cylinder(h = 508, r = 5.08, center = true);
rotate([90, 0, 0])
translate([18, 15.24, 0])
cylinder(h = 508, r = 5.08, center = true);
}
difference() {
translate([114.3, -146.05, 12.7])
cube([228.6, 12.7, 25.4], center = true);
rotate([90, 0, 0])
translate([215.9, 15.24, 0])
cylinder(h = 508, r = 5.08, center = true);
rotate([90, 0, 0])
translate([18, 15.24, 0])
cylinder(h = 508, r = 5.08, center = true);
}
Console
Compiling design (CSG Tree generation)...
Compiling design (CSG Products generation)...
Geometries in cache: 7
Geometry cache size in bytes: 10280
CGAL Polyhedrons in cache: 0
CGAL cache size in bytes: 0
Compiling design (CSG Products normalization)...
Normalized tree has 344 elements!
Compile and preview finished.
Total rendering time: 0:00:00.044
Error-Log
Show All
Group File Line Info
Viewport: translate = [58.44 48.68 27.74], rotate = [57.80 0.00 50.90], distance = 1911.15, fov = 22.50 (1124x1030)